

Techno Link College



Enhance the use of traditional Medicine's Individual assignment

Pharmacy department

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Date: 02/08/2026

I. Need for Extraction (Why Extraction is Necessary)

Extraction is the process of separating the active chemical constituents from crude natural materials (plants, animals, or minerals) using suitable solvents.

1. To isolate active constituents

- Plants contain many compounds.
- Extraction separates the medicinal compounds from inactive materials.

Example: Extracting caffeine from coffee beans.

2. To improve purity

- Removes unwanted substances such as:
 - Cellulose
 - Wax
 - Chlorophyll
 - Dirt
 - Fibers

Result: A cleaner medicinal product.

3. To prepare pharmaceutical products

Extracts are used in manufacturing:

- Tablets
- Capsules
- Syrups
- Tinctures
- Creams
- Ointments

4. To increase therapeutic effectiveness

Concentrating active ingredients makes medicines more effective.

5. To preserve active compounds

Extraction helps protect active ingredients from decomposition during storage.

6. To standardize herbal medicines

Extraction allows manufacturers to produce products containing consistent amounts of active constituents.

7. To facilitate chemical analysis

Researchers extract compounds before:

- Identification
- Purification
- Quality control
- Drug discovery

8. To improve dosage accuracy

Extracts contain measurable concentrations of active ingredients, allowing accurate dosing.

II. Advantages and Disadvantages of Extraction Methods

A. Maceration

Definition: Plant material is soaked in solvent at room temperature.

Advantages

- Simple method
- Low cost
- No heating required
- Suitable for heat-sensitive compounds

Disadvantages

- Takes a long time
- Low extraction efficiency
- Requires large volumes of solvent
- Risk of microbial growth

B. Infusion

Definition: Hot or cold water is poured over plant material for a short period (similar to making tea).

Advantages

- Easy and fast
- Cheap
- Suitable for soft plant parts
- Safe for household preparation

Disadvantages

- Short shelf life
- Water extracts only water-soluble compounds
- Not suitable for hard materials

C. Decoction

Definition: Plant material is boiled in water.

Advantages

- Good for hard plant materials
- Extracts compounds efficiently
- Simple procedure

Disadvantages

- Heat destroys heat-sensitive constituents
- Volatile oils may evaporate
- Limited to water-soluble compounds

D. Percolation

Definition: Solvent slowly passes through powdered plant material.

Advantages

- High extraction efficiency
- Faster than maceration
- Produces concentrated extract
- Suitable for large-scale production

Disadvantages

- Requires special equipment
- More expensive
- Fine powders may block solvent flow

E. Soxhlet Extraction

Definition: Solvent continuously circulates through the plant material under heating.

Advantages

- Efficient extraction
- Less solvent required
- Automatic continuous process
- High yield

Disadvantages

- Not suitable for heat-sensitive compounds
- High energy consumption
- Time-consuming
- Requires laboratory equipment

F. Steam Distillation

Definition: Steam is used to extract volatile oils.

Advantages

- Produces pure essential oils
- No organic solvents required
- Suitable for aromatic plants

Disadvantages

- Only works for volatile compounds
- Expensive equipment
- Heat may affect sensitive constituents

G. Cold Expression (Expression)

Definition: Mechanical pressing without heat, mainly used for citrus peels.

Advantages

- No heat damage
- Natural product quality maintained
- Simple process

Disadvantages

- Only suitable for oil-rich materials
- Lower yield than solvent extraction

H. Ultrasound-Assisted Extraction (UAE)

Advantages

- Fast extraction
- High yield
- Less solvent required
- Energy efficient

Disadvantages

- Expensive equipment
- Not suitable for every plant material

I. Microwave-Assisted Extraction (MAE)

Advantages

- Very rapid
- High extraction efficiency
- Reduced solvent consumption

Disadvantages

- High equipment cost
- Heat may damage sensitive compounds